SAI ANJALI GADE

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PROFILE

Health Data Science graduate student with a background in Biomedical Engineering and 2+ years of analytics experience. Skilled in Python, SQL, R, and machine learning with a strong foundation in data cleaning, modeling, and visualization. Experienced in translating complex data into actionable insights to support business decisions and client deliverables. Passionate about leveraging data science to improve healthcare outcomes and operational efficiency.

WORK EXPERIENCE

Saint Louis University — St. Louis, MO, US
Capstone Experience — June 2025 – August 2025

Who Is Falling Behind? Analysis of Chronic Respiratory Disease Mortality, Tobacco Use, Air Pollution, and GDP per Capita in Middle-Income Americas (2000–2021)| R, Python, Data Analysis, Machine Learning, ggplot2, dplyr, scikit-learn

- Integrated WHO, PAHO, and World Bank datasets to analyze CRD mortality trends in 20+ countries, evaluating economic, environmental, and behavioral risk factors.
- Applied OLS, Ridge, and Lasso regression, achieving R^2 of 0.999, and identified aging and COPD as key mortality predictors.
- Built Random Forest models with 79% classification accuracy and near-perfect regression fit to capture non-linear relationships.
- Conducted k-means clustering to classify nations as “On Track,” “At Risk,” or “Off Track” for WHO 2030 NCD targets, enabling targeted policy recommendations.
- Designed more than 15 advanced visualizations (heat maps, choropleth maps, time-series) to present gender disparities, economic-health patterns, and priority intervention areas.

Cognizant Technologies Solutions — Hyderabad, Telangana, India
Programmer Analyst — August 2022 – July 2023

- Collaborated with cross-functional teams to conduct data-driven root cause analysis for over 100 server-side code deployments, improving software reliability and reducing defects by 15%.
- Built and maintained system logic flowcharts and process diagrams to support technical documentation and business process understanding.
- Supported internal users by developing and refining data tools and system reports, enabling over 50 employees to complete operational tasks efficiently and enhancing business operations.
- Performed regression and system-level testing to ensure data integrity and 98% software accuracy prior to production releases, demonstrating accountability for overall solution quality.
- Engaged in regular reporting and data validation to monitor the effectiveness of updates and inform continuous improvement strategies.

Cognizant Technologies Solutions — Hyderabad, Telangana, India
Programmer Analyst Intern — April 2022 – July 2022

- Assisted with identifying and resolving over 20 technical issues by analyzing system data and performance logs in collaboration with senior engineers.
- Organized and maintained project documentation, spreadsheets, and over 500 tracked files, improving data accessibility and version control.
- Participated in technical workshops and sessions focused on data workflows, development best practices, and software tools, demonstrating a proactive approach to learning new domains.

PROJECTS

Influence of Medical History and Lifestyle on Premiums R, RStudio, Data Analysis, ggplot2, dplyr

- Conducted an in-depth analysis of over 1,000 health insurance records to explore how personal health factors influence insurance premiums, providing insights into cost-driving variables.
- Utilized statistical methods (correlation analysis, ANOVA, multiple linear regression) to measure the impact of age, BMI, smoking status, and exercise frequency on insurance costs.
- Developed predictive models explaining over 65% of the variation in insurance costs, highlighting key cost-driving variables for policy development.
- Designed visual summaries and plots to present findings clearly for reporting and policy development purposes.

Hospital Annual Utilization Analysis Python, Data Handling, Machine Learning, Dask, Data Visualization

- Worked on a large-scale dataset (over 500,000 hospital records) to assess resource usage patterns and operational performance, identifying areas for efficiency improvements.
- Performed extensive data cleaning and transformation to ensure high-quality inputs for analysis and modeling.
- Applied Dask for parallel computing, increasing data processing efficiency by 30%.
- Used genetic algorithms for feature selection, improving prediction model accuracy by 12% and supporting better hospital resource planning.

Predicting Stroke Risk Based on Health and Lifestyle Factors Python, Machine Learning Models

- Utilized Python (scikit-learn, Pandas, Matplotlib, Seaborn) to build and evaluate machine learning models (SVM, Random Forest, KNN, Naive Bayes) for stroke risk prediction.
- Analyzed data from 5,110 patients, achieving 94% accuracy with SVM (RBF kernel) in predicting stroke risk.
- Compared various machine learning models, with SVM and KNN showing highest precision (>93%) for the majority class.
- Evaluated SVM kernels and identified age, hypertension, and glucose as top predictors, providing actionable insights for risk assessment.
- Addressed class imbalance (<5% stroke cases) and recommended rebalancing strategies to improve minority class recall.

SKILLS

Programming & Data Languages: SQL (Strong), Python, R, C, Java

Data Analysis & Modeling: Statistical Analysis, Data Cleaning, Predictive Modeling, Data Visualization, Clinical Data Interpretation, Public Health Analytics, Outcomes Evaluation, Healthcare Reporting Standards, Requirements Elicitation, Business Process Understanding, Unit/UAT Testing

Tools & Technologies: Power BI, Tableau, Jupyter, Snowflake, MATLAB, Simulink, Microsoft Office Suite (Outlook, Word, PowerPoint, Excel)

Machine Learning: SVM, Random Forest, KNN, Naive Bayes, Feature Selection, Model Evaluation

EDUCATION

Master of Science in Health Data Science August 2023 – August 2025
*SAINT LOUIS UNIVERSITY [GPA: **3.71**]* Saint Louis, Missouri

- Emphasis on Data Analytics, Data Management, AI and Machine Learning, SQL, Python, R.

Bachelor of Technology in Biomedical Engineering June 2018 – May 2022
*VIGNAN'S UNIVERSITY [GPA: **8.61**]* Andhra Pradesh, India

- Focus on Biomedical Equipment, MATLAB, Image Processing, Python, Medical Signal Processing.

CERTIFICATIONS

Google Analytics: April 2025 – April 2026

Human Research: Biomedical Research Investigators and Key Personnel, CITI Program: May 2024 – May 2027

Acute Ischemic Stroke Evaluation & Management, Cleveland Clinic Center for Continuing Education: Feb 2025